

SYDNEY GRAMMAR SCHOOL



2014

FORM V
EXAMINATION

Physics

8:40 am Tuesday 20 MAY

Working Time: 2 hours

General Instructions

- Working time – 2 hours
- Board-approved calculators may be used
- Write using blue or black pen
- Draw diagrams using pencil
- A Data Sheet, Formula Sheet and Periodic Table are provided at the back of this paper
- Write your name at the top of the Multiple Choice Answer Sheet & all pages of Part B.
- Hand in your Multiple Choice Sheet and all of Part B in one bundle. (Do not staple together)

Total marks (95)

This paper has three parts: Part A and Part B

Part A

Total marks (10)

- Attempt ALL Questions
- Allow about 15 minutes for this Part

Part B

Total marks (85)

- Attempt ALL Questions
- Allow about 1 hour 45 minutes for this Part

CHECKLIST

Each boy should have the following:

1 Question Paper

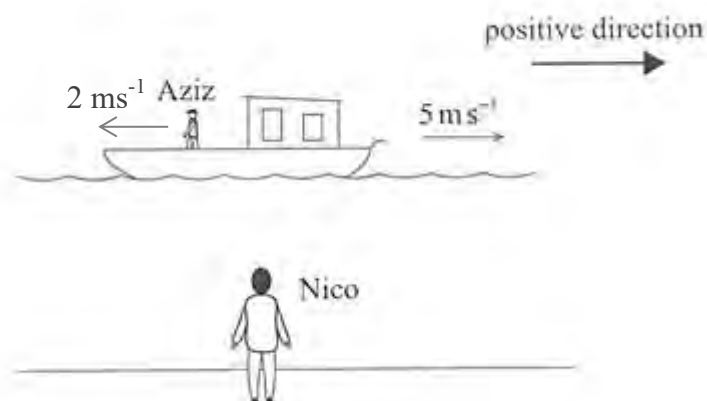
1 Multiple Choice Answer Sheet

1 – PCK	2 – MTK	3 – AAH	4 – SRW
5 – AAH	6 – MRW	7 – PCK	

EXAMINERS: MRW/AAH/PCK/SRW/MTK

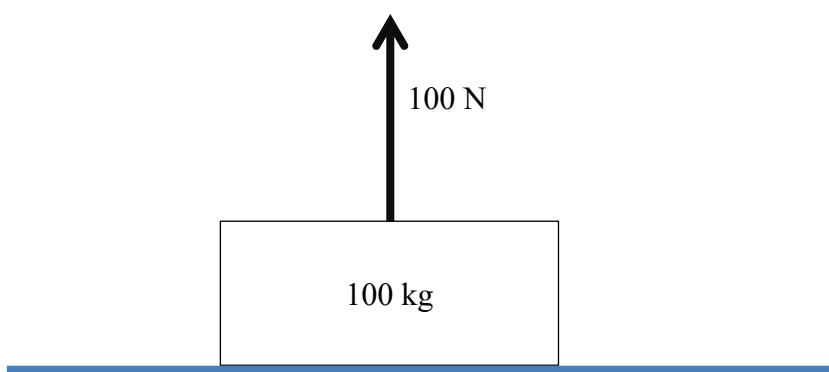
- 1 Which of the following lists contains three vector quantities?
- (A) Momentum, energy, impulse
 - (B) Force, speed, displacement
 - (C) Velocity, acceleration, time
 - (D) Acceleration, force, momentum
- 2 Which is the correct unit for impulse?
- (A) Newton per second
 - (B) Newton second
 - (C) Kilogram metre per second squared
 - (D) Kilogram per metre per second
- 3 A car travelling at 60 kmh^{-1} then applies maximum braking and comes to a stop in 10 m.
With the same braking force applied, what is the new stopping distance for the same car when travelling at 120 kmh^{-1} ?
- (A) 10 m
 - (B) 20 m
 - (C) 30 m
 - (D) 40 m
- 4 An object, initially at rest, travels a distance d in a total time t at a constant acceleration.
What is the total time taken for the object to travel $16d$ from rest at the same acceleration?
- (A) $2t$
 - (B) $4t$
 - (C) $8t$
 - (D) $16t$
- 5 A 100 g tennis ball hits the ground at 2 ms^{-1} and rebounds upwards at 1.4 ms^{-1} .
What is the change in momentum of the tennis ball?
- (A) 0.34 kgms^{-1} upwards
 - (B) 0.34 kgms^{-1} downwards
 - (C) 0.06 kgms^{-1} upwards
 - (D) 0.06 kgms^{-1} downwards

- 6 A boat is moving in the direction shown below with a constant speed of 5 ms^{-1} as measured by Nico who is at rest on the bank of the river. Aziz walks along the deck of the boat in the direction shown with a constant speed of 2 ms^{-1} measured relative to the boat.



If the positive direction is as shown in the above diagram, what is the velocity of Aziz according to Nico?

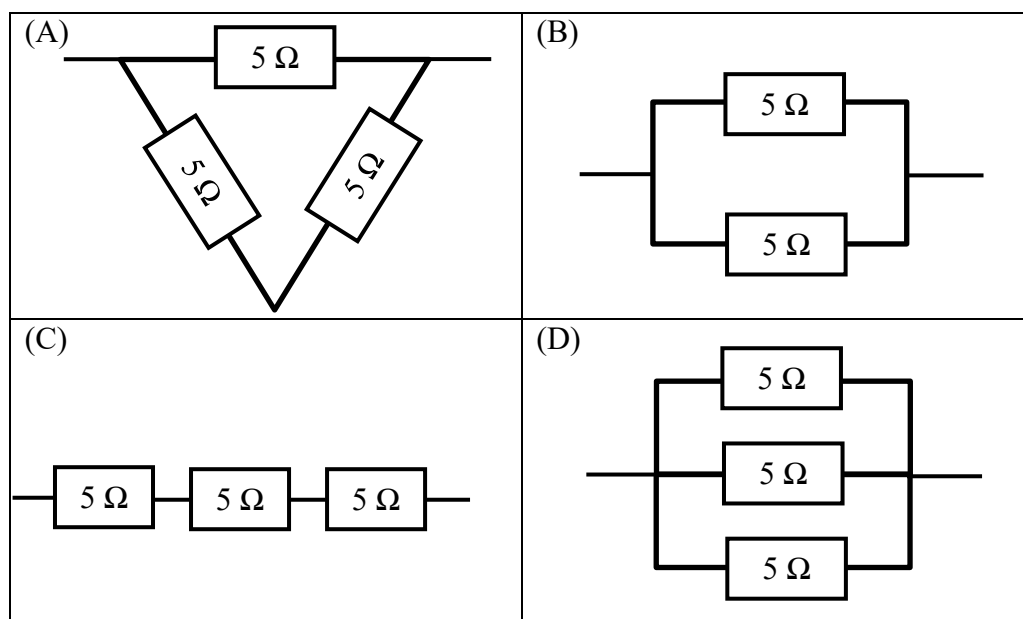
- (A) $+3 \text{ ms}^{-1}$
(B) -3 ms^{-1}
(C) -7 ms^{-1}
(D) $+7 \text{ ms}^{-1}$
- 7 A rope with a tension of 100 N supports a 100 kg block resting on the ground as shown below.



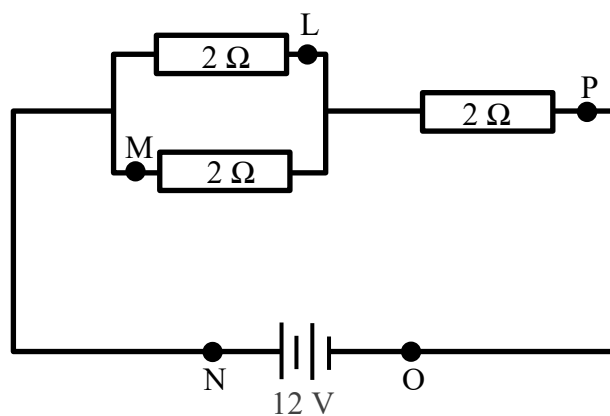
What is the magnitude of the force of the ground up on the block?

- (A) 0 N
(B) 100 N
(C) 880 N
(D) 980 N

- 8 Which of the following arrangements of $5\ \Omega$ resistors has the least total resistance?



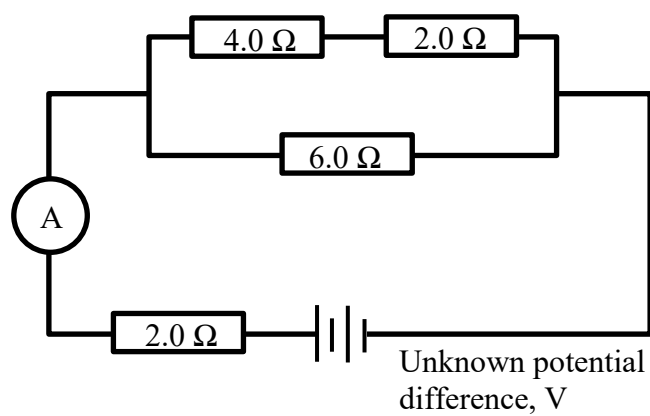
- 9 Consider the following circuit diagram.



Between which two points (L, M, N, O, P) is the potential difference 4 V?

- (A) N and M
- (B) L and M
- (C) L and P
- (D) P and O

10 Consider the following circuit diagram.



The ammeter reads 4.0 A.

What is the current flowing through the 6.0 Ω resistor and the value of the unknown potential difference of the power supply, V?

	current in 6.0 Ω resistor (A)	potential difference (V)
(A)	1.0	10
(B)	2.0	10
(C)	2.0	20
(D)	1.0	20

Part B**Total marks (85)****Attempt ALL Questions****Allow about 1 hour and 45 minutes for this Part**

Class

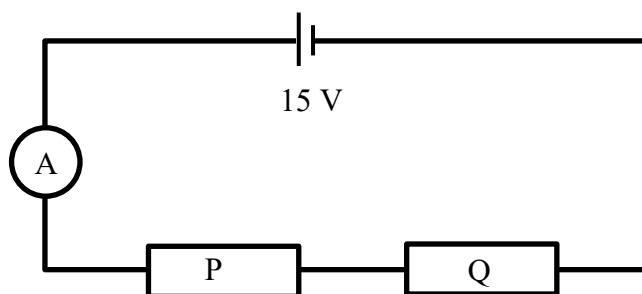
Name

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Question 11 (4 marks)**Marks**

Two resistors, labelled P and Q, are connected in a circuit as shown below.



When a steady current of 3 A flows in the circuit, the potential difference across resistor Q is 9 V.

- (a) Calculate the potential difference across resistor P.

1

- (b) Calculate the resistance of resistor P.

1**Question 11 continued on next page.**

Class

Name

Question 11 continued.**Marks**

- (c) Calculate the total charge leaving the battery in one minute when the ammeter reads 3 A.

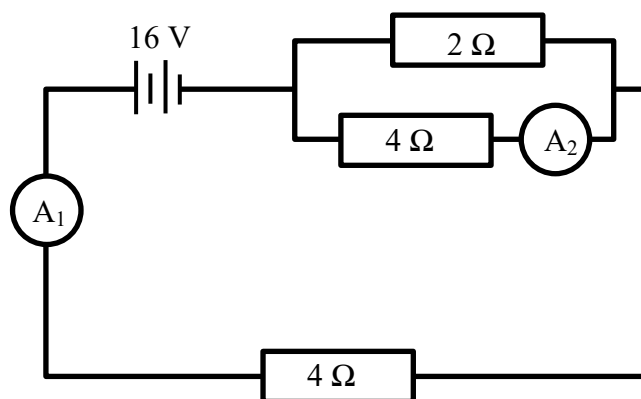
2

ClassName

Marks

Question 12 (5 marks)

This question refers to the circuit diagram below.



- (a) Determine the combined resistance of the two parallel resistors.

2

- (b) What current will be measured by the ammeter A₁?

2

- (c) What current will be measured by the ammeter A₂?

1

Class

Name

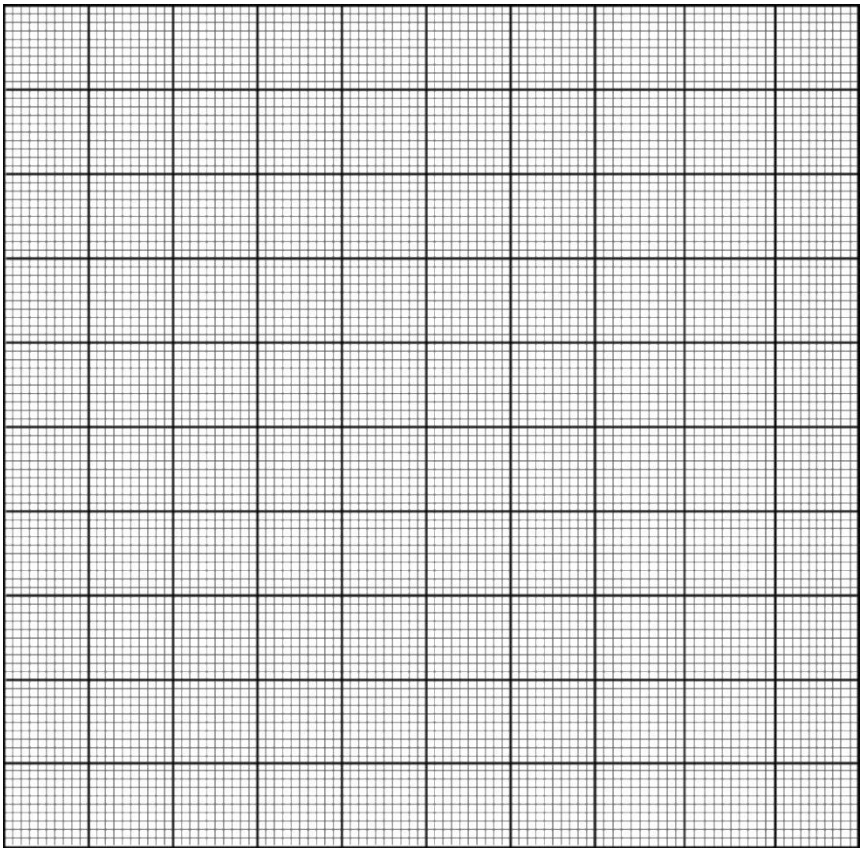
Question 13 (5 marks)

Marks

The data in the following table shows how the potential difference (V) across a resistor varies as the current (I) through it is altered.

current (A)	potential difference (V)
0.0	0.0
0.4	1.5
0.8	3.0
1.2	4.5
1.6	6.0
2.0	7.5

- (a) Plot a graph of potential difference (on the y axis) versus current for the resistor.



Question 13 continued on next page.

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Class

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Name

Question 13 continued.**Marks**

- (b) Using the graph, or otherwise, calculate the resistance of the resistor.

_____	1

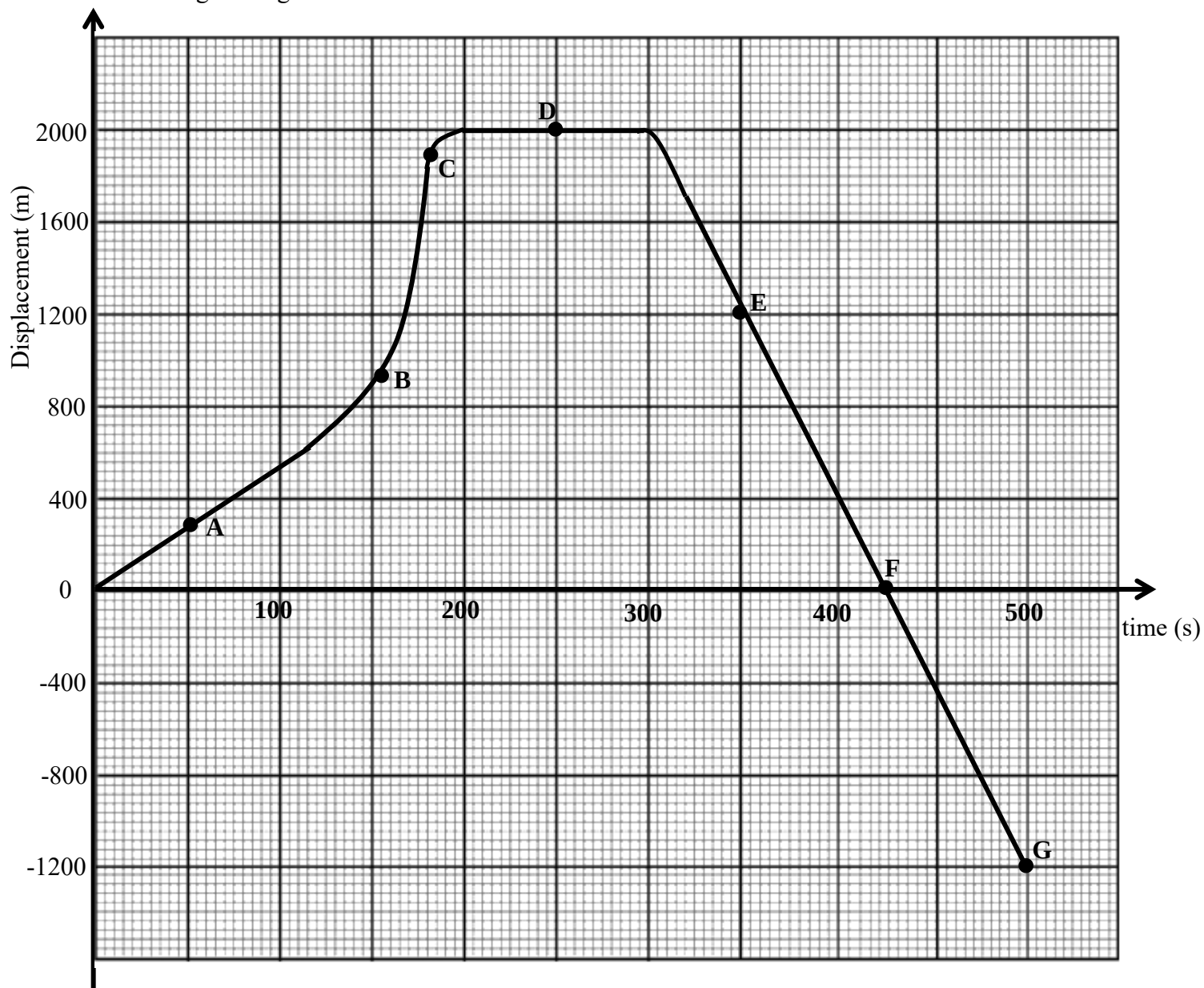
Class

Name

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ClassName**Question 14** (7 marks)**Marks**

The displacement versus time graph shown below represents the motion of a train along a straight track.



Question 14 continued on next page.

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Class

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Name

Question 14 continued.**Marks**

(a) Use the letters on the graph to identify one point where:

i. the train is at rest.

1

ii. the train is accelerating.

1

iii. the train is back at the original position.

1

iv. the train has maximum speed.

1

Question 14 continued on next page.

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Class

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Name

Question 14 continued.**Marks**

(b) Determine:

i. the total distance travelled by the train for the 500 s.

1

ii. the magnitude of the average velocity of the train for the 500 s.

2

Class

Name

Question 15 (6 marks)**Marks**

In 1997, a high speed car achieved the World Land Speed Record and became the first car to officially break the sound barrier. The car accelerated uniformly in two stages as shown in the table below. The car started from rest.

Stage	Time (s)	Speed attained at end of stage (ms ⁻¹)
1	0.0 – 4.0	44.0
2	4.0 – 12.0	340

- (a) Calculate the acceleration of the car in Stage 1.

1

- (b) Calculate the total distance travelled by the car in 12.0 s.

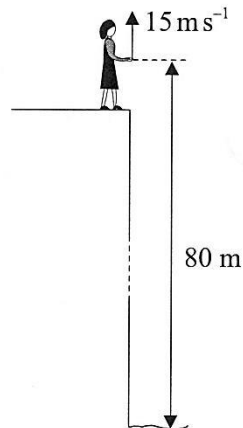
4

- (c) Calculate the average speed of the car over the 12.0 s.

1

ClassName**Question 16** (5 marks)**Marks**

Sally stands on the edge of a vertical cliff and throws a stone vertically upwards.



The stone leaves her hand with a speed of 15 ms^{-1} at the instant her hand is 80 m above the surface of the sea. Air resistance is negligible.

- (a) Calculate the maximum height reached by the stone as measured from the point where it is thrown.

2

- (b) Determine the time for the stone to reach the surface of the sea after leaving Sally's hand.

3

Class

Name

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Class

Name

Question 17 (5 marks)**Marks**

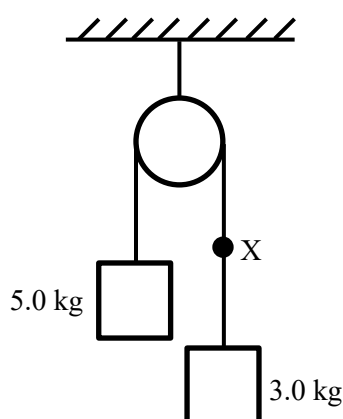
A plane at Airport A needs to fly to Airport B, which is 270 km North, in 30 minutes. At the time of the flight, there is a constant wind blowing at 43 ms^{-1} East.

By drawing a vector diagram, determine the direction of flight of the plane and the speed of the plane through the air.

5

ClassName**Question 18** (4 marks)**Marks**

A simple vertical Atwood machine consists of a 3.0 kg mass and a 5.0 kg mass supported from a frictionless pulley by a light inextensible string as shown in the diagram below.



- (a) Calculate the magnitude of the acceleration of the two masses.

2

- (b) Calculate the magnitude of the tension force in the string at X.

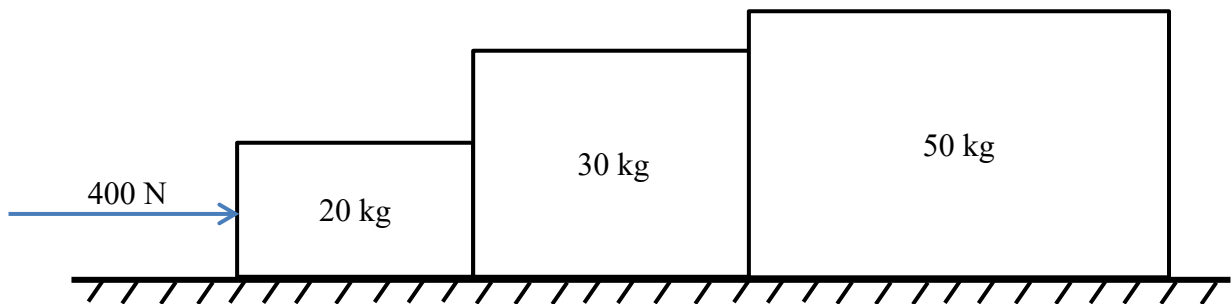
2

Class

Name

Question 19 (4 marks)**Marks**

Three blocks are pushed to the right on a smooth (ie frictionless) horizontal surface as shown below.



- (a) Calculate the magnitude of the acceleration of the masses.

1

- (b) Determine the magnitude of the net force on the 30 kg mass.

1

- (c) Calculate the magnitude of the force of the 30 kg mass on the 20 kg mass.

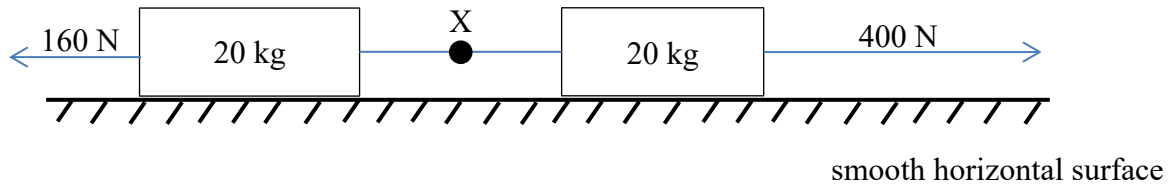
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Class

Name

Marks**Question 20** (4 marks)

Two masses are pulled along a smooth, horizontal surface by two horizontal forces as shown below.



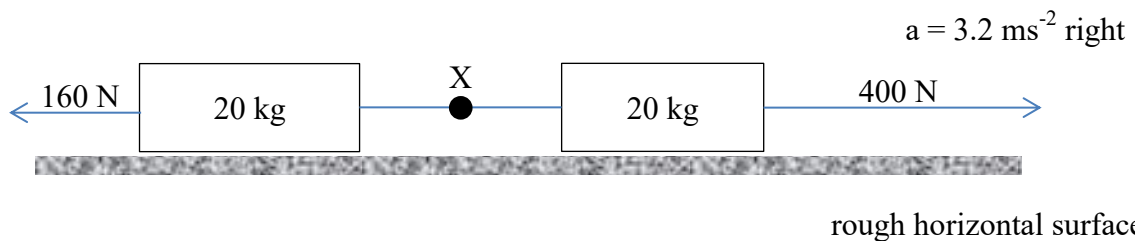
- (a) Determine the magnitude of the acceleration of the masses.

1

- (b) Determine the magnitude of the tension force in the rope at X.

1

- (c) If the same arrangement of masses is now placed on a rough surface, it is found that the acceleration is 3.2 ms^{-2} to the right as shown in the following diagram.



Calculate the magnitude of the force of friction acting on each mass (assuming that it is the same for each mass).

2

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Class

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Name

Marks**Question 21** (2 marks)

A man of mass 80 kg is climbing up a rope as shown below.



If the acceleration of the man is 1.0 ms^{-2} upwards, determine the magnitude of the tension force in the rope.

2

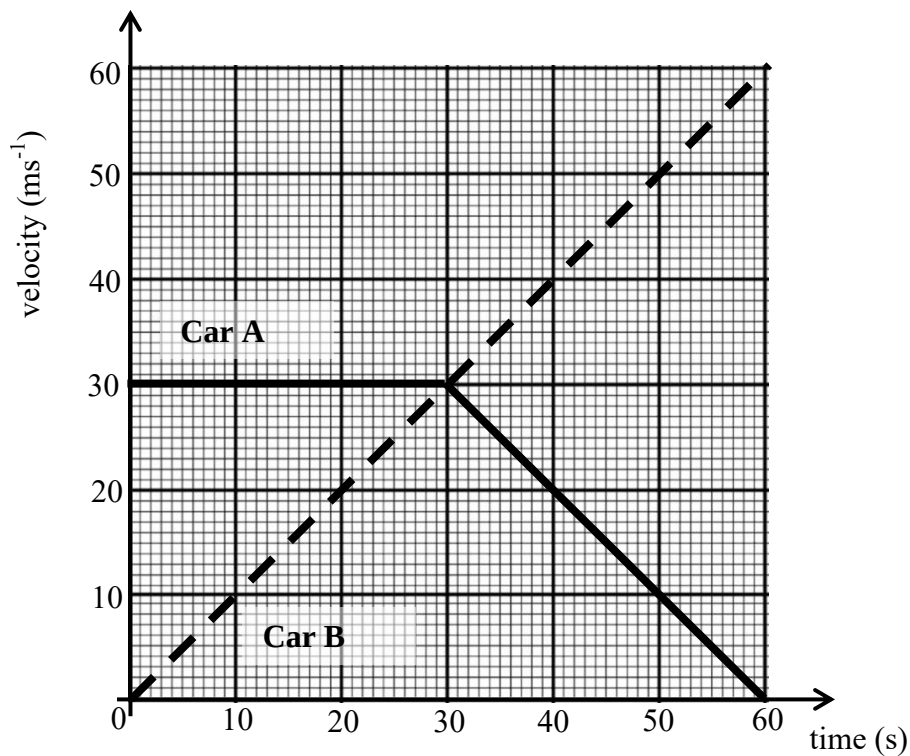
Class

Name

Marks

Question 22 (5 marks)

The graph shown below represents the velocity of two cars racing along a straight road over a period of 60 s. The race begins at $t = 0$ s when the two cars are side by side.



- (a) Determine the speed of car A at 10 s.

1

- (b) Calculate the magnitude of the acceleration of car B at 20 s.

1

Question 22 continued on next page.

Class

Name

Question 22 continued.**Marks**

- (c) Calculate the time when car B catches up with Car A again.

3

ClassName**Question 23** (7 marks)**Marks**

On a distant planet, an alien pupil performs an experiment to measure the acceleration due to gravity. He drops (i.e. from rest) a ball from various heights and measures the time the ball takes to hit the ground.

The equation relating the height dropped by the ball from rest and time (t) is given by:

$$height = \frac{1}{2}gt^2$$

where g = acceleration due to gravity.

He obtains the following data:

Height (m)	Time (s)	
0.5	0.24	
1.0	0.41	
1.5	0.45	
2.0	0.55	
2.5	0.61	
3.0	0.65	

Question 23 continued on next page.

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Class

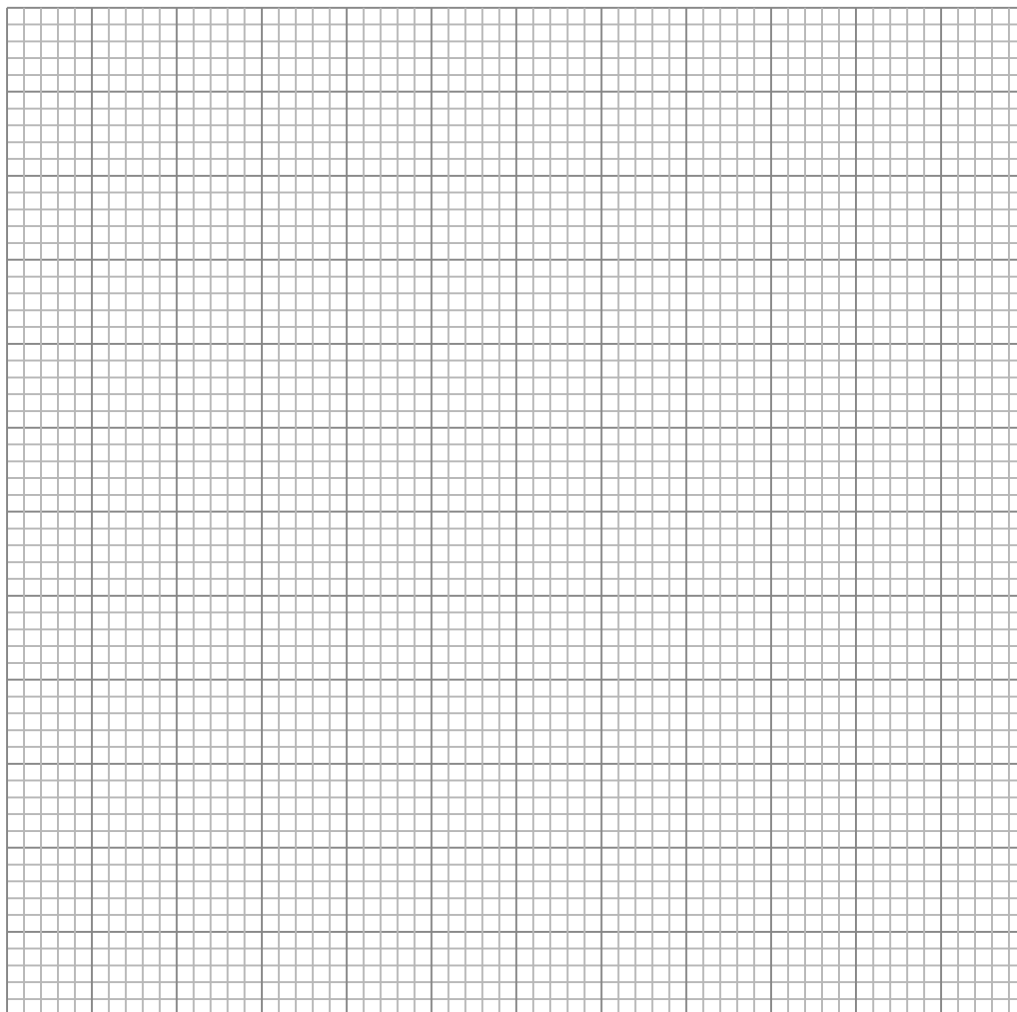
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Name

Marks**Question 23 continued.**

- (a) Plot a ***straight-line graph*** of the pupil's data that will allow you to determine the acceleration due to gravity on the planet.

(Note: you may need to manipulate the data you are given in the table – an extra column has been provided for you to do so.)

**4****Question 23 continued on next page.**

Class

Name

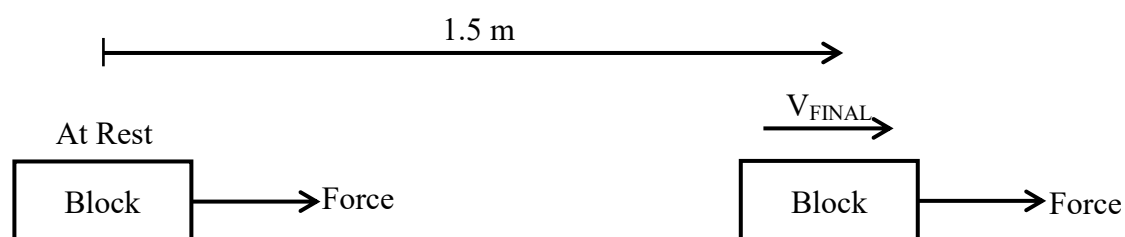
Question 23 continued.**Marks**

- (b) Calculate the value for the acceleration due to gravity obtained by the alien pupil.

3

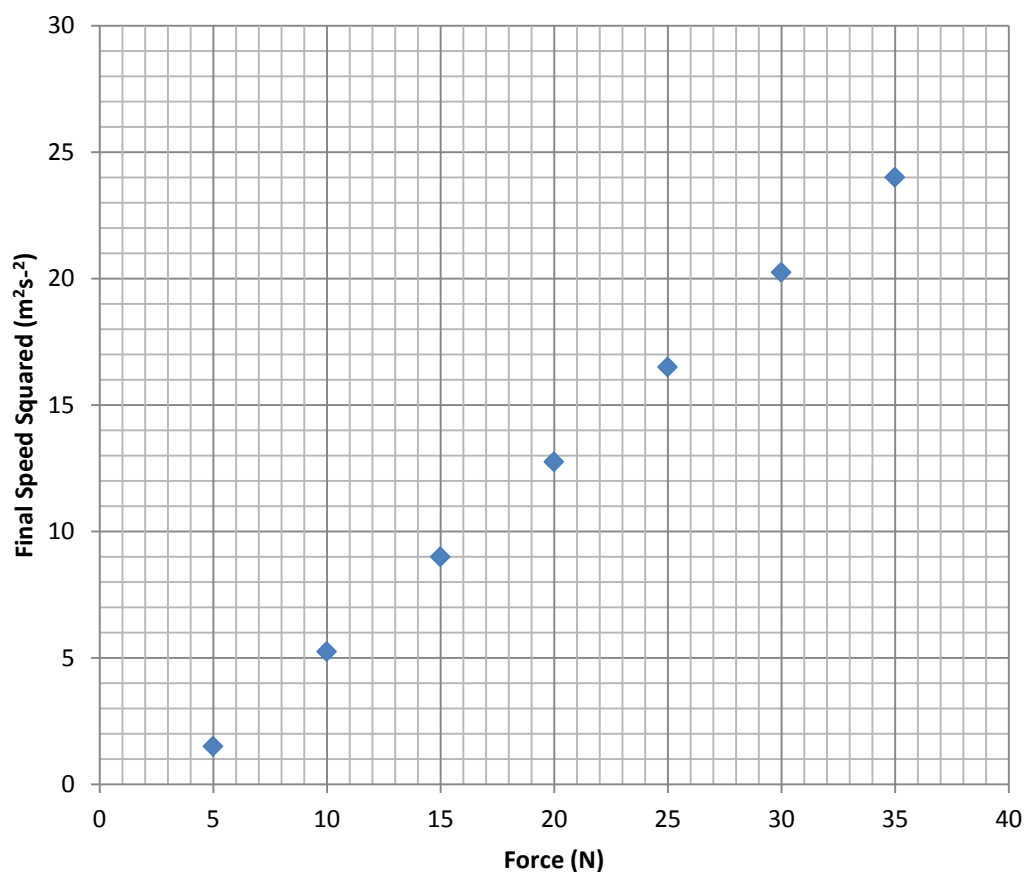
ClassName**Question 24** (3 marks)**Marks**

The diagram below shows an experiment performed by a boy in a Physics lesson.



The boy pulls a block, from rest, along the desk over a distance of 1.5 m with a constant force. A light gate measures the instantaneous speed of the block as it passes the 1.5 m mark.

The boy believes that a constant force of friction acts on the block during his experiments. He varies the force with which he pulls the block and obtains the following graph:



Question 24 continued on next page.

Class

Name

Question 24 continued.**Marks**

Use the graph to determine:

- (a) the force of friction acting.

1

- (b) the mass of the block.

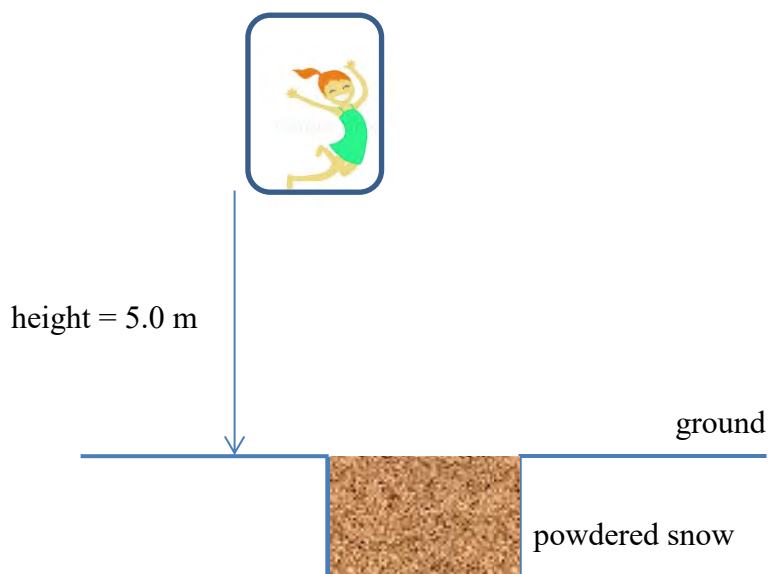
2

Class

Name

Question 25 (6 marks)**Marks**

Natalia, of mass 50 kg, drops from a window 5.0 m above the ground as shown in the following diagram.



- (a) Determine Natalia's gravitational potential energy when she is 5.0 m above the ground.

1

- (b) Determine the magnitude of her velocity when she first hits the ground.

2

Question 25 continued on next page.

Class

Name

Question 25 continued.**Marks**

- (c) The ground onto which Natalia falls is made of powdered snow. The snow applies a constant braking force upwards of $5.20 \times 10^3 \text{ N}$ on her.

Calculate the maximum distance Natalia will sink into the snow before stopping.

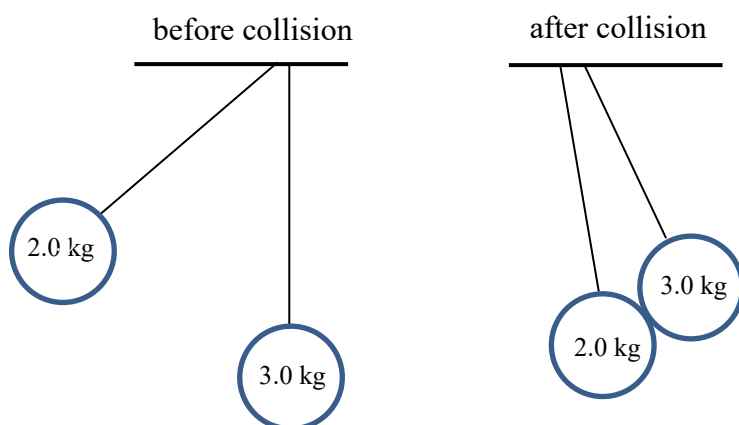
3

Class

Name

Question 26 (5 marks)**Marks**

A 2.0 kg pendulum moving at 4.0 ms^{-1} collides with a stationary 3.0 kg pendulum and then they stick together.



- (a) Determine the magnitude of the momentum of the 2.0 kg pendulum immediately before the collision.

1

- (b) Determine the velocity of the combined pendulums immediately after the collision.

2

Question 26 continued on next page.

Class

Name

Question 26 continued.**Marks**

- (c) Ignoring the diameters of the pendulums, determine the vertical height the combined pendulums reach after the collision.

2

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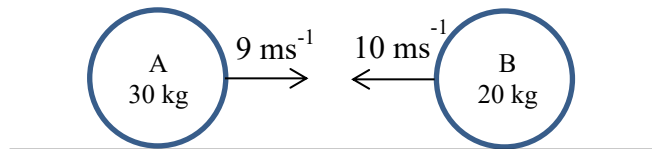
Class

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Name

Marks**Question 27** (8 marks)

Two objects, A and B, approach each other and then collide. After the collision object A is travelling at half the **speed** of object B.



- (a) Calculate the total momentum of this system before the collision.

2

- (b) (i) Find all possible solutions for the final velocities of object A and object B.

4

- (ii) Identify which solution you think is most likely. Justify your answer.

2

Class

Name

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Physics

Data Sheet

Charge on the electron, q_e	$-1.602 \times 10^{-19} \text{ C}$
Mass of electron, m_e	$9.109 \times 10^{-31} \text{ kg}$
Mass of neutron, m_n	$1.675 \times 10^{-27} \text{ kg}$
Mass of proton, m_p	$1.673 \times 10^{-27} \text{ kg}$
Speed of sound in air	340 m s^{-1}
Earth's gravitational acceleration, g	9.8 m s^{-2}
Radius of Earth, R_E	$6.4 \times 10^6 \text{ m}$
Speed of light, c	$3.00 \times 10^8 \text{ m s}^{-1}$
Magnetic force constant, $\left(k \equiv \frac{\mu_0}{2\pi} \right)$	$2 \times 10^{-7} \text{ N A}^{-2}$
Universal gravitational constant, G	$6.7 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Mass of Earth	$6.0 \times 10^{24} \text{ kg}$
Planck's constant, h	$6.626 \times 10^{-34} \text{ J s}$
Rydberg's constant, R (hydrogen)	$1.097 \times 10^7 \text{ m}^{-1}$
Atomic mass unit, u	$1.661 \times 10^{-27} \text{ kg}$ 931.5 MeV/c^2
1 eV	$1.602 \times 10^{-19} \text{ J}$
Density of water, ρ	$1.00 \times 10^3 \text{ kg m}^{-3}$
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

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FORMULAE SHEET FORM V ONLY

$$v_{av} = \frac{\Delta r}{\Delta t}$$

$$a_{av} = \frac{\Delta v}{\Delta t} = \frac{v-u}{t}$$

$$v = u + at$$

$$v^2 = u^2 + 2ar$$

$$r = ut + \frac{1}{2}at^2$$

$$\Sigma F = ma$$

$$F = \frac{mv^2}{r}$$

$$E_k = \frac{1}{2}mv^2$$

$$E_p = mgh$$

$$W = Fr$$

$$p = mv$$

$$\Delta p = F_n t$$

$$F = mg$$

$$E = \frac{F}{q}$$

$$E = \frac{V}{d}$$

$$I = \frac{Q}{t}$$

$$R = \frac{V}{I}$$

$$P = VI$$

$$\text{Energy} = VIt$$

$$v = f\lambda$$

$$f = \frac{1}{T}$$

$$I \propto \frac{1}{d^2}$$

$$\frac{v_1}{v_2} = \frac{\sin i}{\sin r}$$

$$n\lambda = d \sin \theta$$

$$n\lambda = \frac{dx}{L}$$

$$E_p = -\frac{Gm_1m_2}{r}$$

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$$

$$F = \frac{Gm_1m_2}{d^2}$$

$$E = mc^2$$

FORMULAE SHEET

$$\frac{F}{l} = k \frac{I_1 I_2}{d}$$

$$d = \frac{1}{P}$$

$$F = BIl \sin \theta$$

$$M = m - 5 \log \left(\frac{d}{10} \right)$$

$$\tau = Fd$$

$$\frac{I_A}{I_B} = 100(m_B - m_A)/5$$

$$\tau = nBIA \cos \theta$$

$$\frac{V_p}{V_s} = \frac{n_p}{n_s}$$

$$m_1 + m_2 = \frac{4\pi^2 r^3}{GT^2}$$

$$F = qvB \sin \theta$$

$$\frac{1}{\lambda} = R_H \left[\frac{1}{n_f^2} - \frac{1}{n_i^2} \right]$$

$$E = \frac{V}{d}$$

$$\lambda = \frac{h}{mv}$$

$$E = hf$$

$$A_0 = \frac{V_{out}}{V_{in}}$$

$$c = f\lambda$$

$$\frac{V_{out}}{V_{in}} = -\frac{R_f}{R_i}$$

$$Z = \rho v$$

$$\frac{I_r}{I_0} = \frac{[Z_2 - Z_1]^2}{[Z_2 + Z_1]^2}$$

$$\begin{array}{l} \text{Surface area of a} \\ \text{sphere of radius, } R \end{array} = 4\pi R^2$$

PERIODIC TABLE OF THE ELEMENTS

1 H 1.008 Hydrogen		4 Be 9.012 Beryllium										79 Au 197.0 Gold										2 He 4.003 Helium													
3 Li 6.941 Lithium		12 Mg 24.31 Magnesium										Atomic Number Symbol Name										10 Ne 20.18 Neon													
11 Na 22.99 Sodium		20 Ca 40.08 Calcium										Standard Atomic Weight										18 Ar 39.95 Argon													
19 K 39.10 Potassium		21 Sc 44.96 Scandium		22 Ti 47.87 Titanium		23 V 50.94 Vanadium		24 Cr 52.00 Chromium		25 Mn 54.94 Manganese		26 Fe 55.85 Iron		27 Co 58.93 Cobalt		28 Ni 58.69 Nickel		29 Cu 63.55 Copper		30 Zn 65.38 Zinc		31 Ga 69.72 Gallium		32 Ge 72.64 Germanium		33 As 74.92 Arsenic		34 Se 78.96 Selenium		35 Br 79.90 Bromine		36 Kr 83.80 Krypton			
37 Rb 85.47 Rubidium		39 Y 88.91 Yttrium		40 Zr 91.22 Zirconium		41 Nb 92.91 Niobium		42 Mo 95.96 Molybdenum		43 Tc Technetium		44 Ru 101.1 Ruthenium		45 Rh 102.9 Rhodium		46 Pd 106.4 Palladium		47 Ag 107.9 Silver		48 Cd 112.4 Cadmium		49 In 114.8 Indium		50 Sn 118.7 Tin		51 Sb 121.8 Antimony		52 Te 127.6 Tellurium		53 I 126.9 Iodine		54 Xe 131.3 Xenon			
55 Cs 132.9 Caesium		57-71 Lanthanoids		72 Hf 178.5 Hafnium		73 Ta 180.9 Tantalum		74 W 183.9 Tungsten		75 Re 186.2 Rhenium		76 Os 190.2 Osmium		77 Ir 192.2 Iridium		78 Pt 195.1 Platinum		79 Au 197.0 Gold		80 Hg 200.6 Mercury		81 Tl 204.4 Thallium		82 Pb 207.2 Lead		83 Bi 209.0 Bismuth		84 Po Polonium		85 At Astatine		86 Rn Radon			
87 Fr Francium		89-103 Actinoids		104 Rf Rutherfordium		105 Db Dubnium		106 Sg Seaborgium		107 Bh Bohrium		108 Hs Hassium		109 Mt Meitnerium		110 Ds Darmstadtium		111 Rg Roentgenium		112 Cn Copernicium															



Class

Name

PHYSICS Form V May 2014

8:40 am Tues 20 MAY

General Instructions

- Write your class and Candidate Number in the space provided.
- Attempt all questions
- Use a blue or black pen
- Select the alternative A, B, C, or D that best answers the question.
- Fill in the response oval completely.

Multiple Choice ANSWER SHEET

- | | | | | |
|-----|----------------------------------|----------------------------------|----------------------------------|----------------------------------|
| 1. | ☐ A | ☐ B | ☐ C | ☒ |
| 2. | ☐ A | ☒ | ☐ C | ☐ D |
| 3. | ☐ A | ☐ B | ☐ C | ☒ |
| 4. | ☐ A | ☒ | ☐ C | ☐ D |
| 5. | ☒ | ☐ B | ☐ C | ☐ D |
| 6. | ☒ | ☐ B | ☐ C | ☐ D |
| 7. | ☐ A | ☐ B | ☒ | ☐ D |
| 8. | ☐ A | ☐ B | ☐ C | ☒ |
| 9. | ☐ A | ☒ | ☐ C | ☐ D |
| 10. | ☐ A | ☐ B | ☒ | ☐ D |

Mark

Part B**Total marks (85)****Attempt ALL Questions****Allow about 1 hour and 45 minutes for this Part**

Class

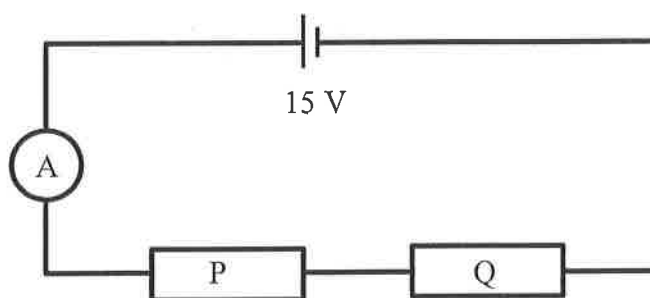
Name

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Question 11 (4 marks)**Marks**

Two resistors, labelled P and Q, are connected in a circuit as shown below.



When a steady current of 3 A flows in the circuit, the potential difference across resistor Q is 9 V.

- (a) Calculate the potential difference across resistor P.

$$V_T = V_P + V_Q$$

$$15 = V_P + 9 \quad \therefore V_P = 6V$$

$$V_P = 15 - 9 \quad \checkmark \text{ (1mk for working)}$$

1

- (b) Calculate the resistance of resistor P.

$$R = V/I = 6/3 = 2\Omega$$

$$[R = \text{Ans(a)}/3] \quad \checkmark \text{ (1mk)}$$

1

Question 11 continued on next page.

Class

Name

Question 11 continued.

Marks

- (c) Calculate the total charge leaving the battery in one minute when the ammeter reads 3 A.

$$Q = It$$

$$= 3 \times 1 \times 60$$

$$= 180 \text{ C}$$

✓ ① mk converting 2
time to
seconds

✓ ① mk correct formula
with $Q = It$
 $= 3 \times 1$

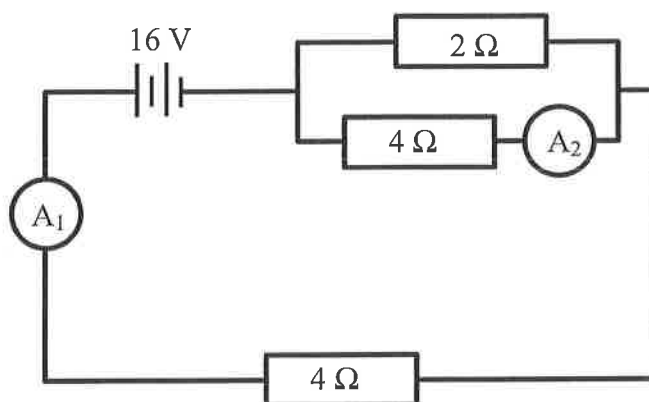
Class

Name

Marks

Question 12 (5 marks)

This question refers to the circuit diagram below.



- (a) Determine the combined resistance of the two parallel resistors.

$$\frac{1}{R} = \frac{1}{4} + \frac{1}{2} = \frac{3}{4} \quad \checkmark \text{ 1mk}$$

$$\therefore R = \frac{4}{3} \Omega = 1.33 \Omega \quad \checkmark \text{ 1mk inverted.}$$

2

- (b) What current will be measured by the ammeter
- A_1
- ?

$$V = IR.$$

$$16 = I \times \left(4 + \frac{4}{3}\right)$$

$$I = 3A.$$

$$\left[I = \frac{16}{\text{total } R} \right] \quad \checkmark \text{ 2mks.}$$

2

- (c) What current will be measured by the ammeter
- A_2
- ?

$$I = \frac{16 - 4 \times 3}{4} = 1A. \quad \checkmark \text{ 1mk.}$$

1

if values
reversed
i.e.
(b) 1A
(c) 3A
then
xx ✓ 1mk.

$$\left[I = \frac{16 - 4 \times \text{Ans(b)}}{4} \right]$$

Note $I = \frac{16}{4} = 4A \quad \times \text{ 0mks}$

Class

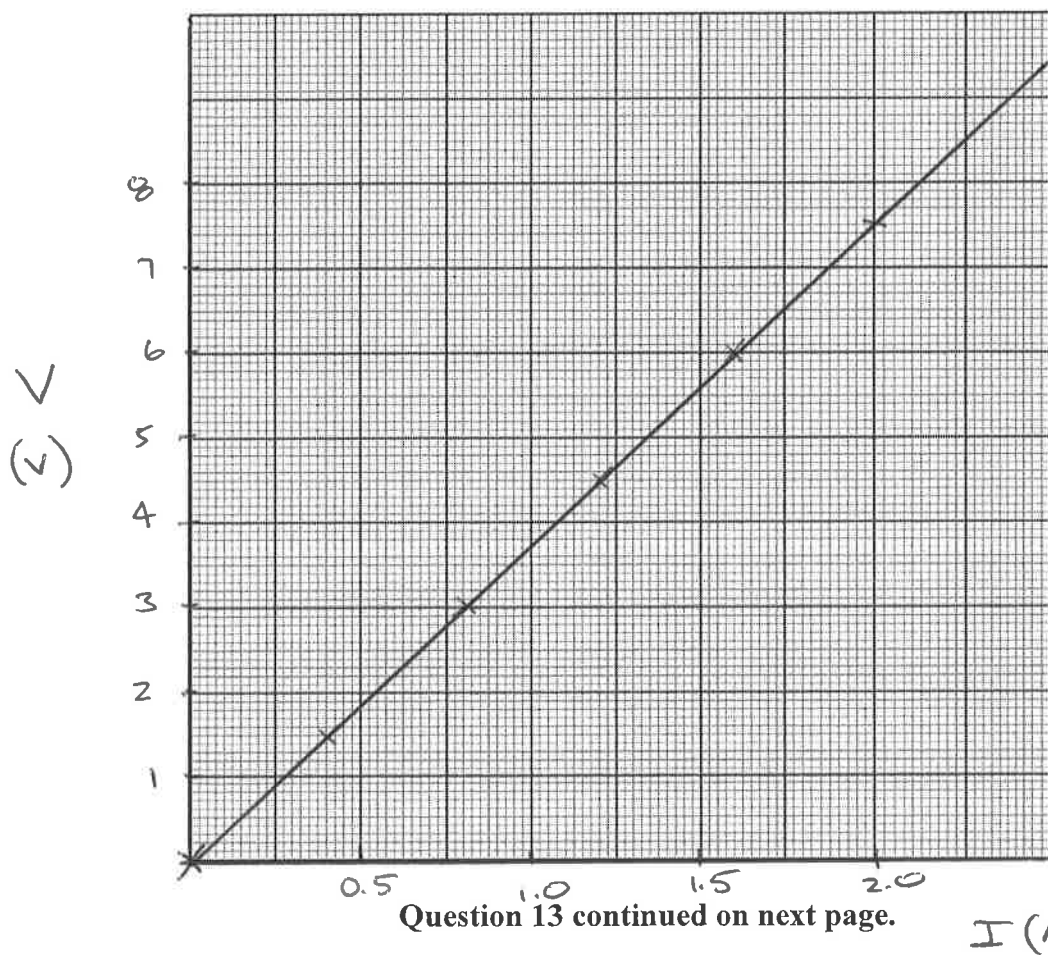
Name

Question 13 (5 marks)**Marks**

The data in the following table shows how the potential difference (V) across a resistor varies as the current (I) through it is altered.

current (A)	potential difference (V)
0.0	0.0
0.4	1.5
0.8	3.0
1.2	4.5
1.6	6.0
2.0	7.5

- (a) Plot a graph of potential difference (on the y axis) versus current for the resistor.



— 1mk if axes wrong way

①mk/ both axes labelled with correct quantities

①mk/ both axes labelled with correct units

①mk/ points plotted correctly + line of best fit drawn.

①mk/ scale chosen is sensible and linear so graph paper is at least 50% used.

Class

Name

Marks

Question 13 continued.

- (b) Using the graph, or otherwise, calculate the resistance of the resistor.

$$R = V/I$$
$$= 7.5/2 = 3.75 \Omega \quad \checkmark \text{ 1mk.}$$

1

CR18.

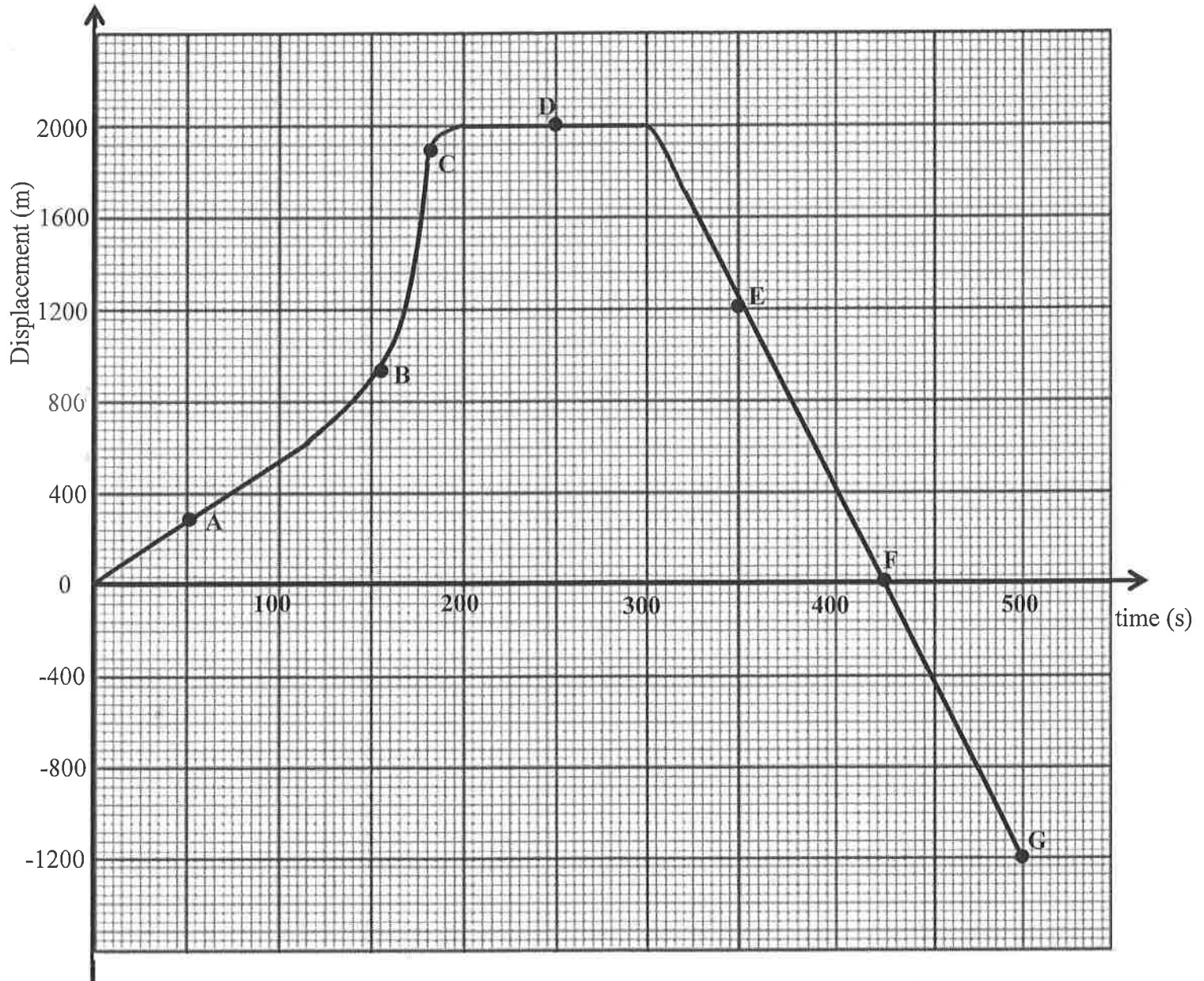
Class

MTK.

Name

Question 14 (7 marks)**Marks**

The displacement versus time graph shown below represents the motion of a train along a straight track.



Question 14 continued on next page.

Class

Name

Question 14 continued.**Marks**

(a) Use the letters on the graph to identify one point where:

i. the train is at rest.

1

D

ii. the train is accelerating.

1

B

iii. the train is back at the original position.

1

F

iv. the train has maximum speed.

1

C.

Question 14 continued on next page.

Class

Name

Question 14 continued.

Marks

(b) Determine:

- i. the total distance travelled by the train for the 500 s.

1

$$r = 2000 + 3200$$
$$= 5,200 \text{ m}$$

- ii. the
- magnitude
- of the average velocity of the train for the 500 s.

2

$$v_{av} = \frac{\Delta \vec{r}}{\Delta t} = \frac{5200}{500}$$
$$= 10.4 \text{ m s}^{-1}$$

Class

 Name
Question 15 (6 marks)**Marks**

In 1997, a high speed car achieved the World Land Speed Record and became the first car to officially break the sound barrier. The car accelerated uniformly in two stages as shown in the table below. The car started from rest.

Stage	Time (s)	Speed attained at end of stage (ms ⁻¹)
1	0.0 – 4.0	44.0
2	4.0 – 12.0	340

- (a) Calculate the acceleration of the car in Stage 1.

$$\vec{a}_1 = \frac{\vec{v} - \vec{u}}{t} = \frac{44 - 0}{4} = 11 \text{ ms}^{-2}$$

1

- (b) Calculate the total distance travelled by the car in 12.0 s.

$$(1) \quad \vec{r} = \vec{u}t + \frac{1}{2}\vec{a}t^2 = 0 + \frac{1}{2} \times 11 \times 4^2 = 88 \text{ m.}$$

4

$$(1) \quad a_2 = \frac{340 - 44}{8} = 37. \text{ ms}^{-2}$$

$$(1) \quad r_2 = 44 \times 8 + \frac{1}{2} \times 37 \times 8^2 = 1536 \text{ m.}$$

$$(1) \quad \text{Total distance} = 1536 + 88 = 1624 \text{ m.}$$

- (c) Calculate the average speed of the car over the 12.0 s.

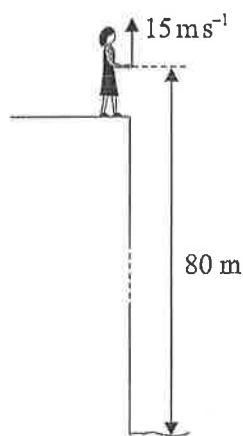
$$V_{AV} = \frac{\Delta r}{\Delta t} = 1624 / 12 = 135.3 \text{ ms}^{-1}$$

1

Class

 Name
Question 16 (5 marks)**Marks**

Sally stands on the edge of a vertical cliff and throws a stone vertically upwards.



The stone leaves her hand with a speed of 15 ms^{-1} at the instant her hand is 80 m above the surface of the sea. Air resistance is negligible.

- (a) Calculate the maximum height reached by the stone as measured from the point where it is thrown.

$$v^2 = u^2 + 2ar$$

$$(1) \quad 0 = 15^2 + 2 \times -9.8 \times r$$

$$(1) \quad r = 11.48 \text{ m}$$

2

- (b) Determine the time for the stone to reach the surface of the sea after leaving Sally's hand.

$$r = ut + \frac{1}{2}at^2$$

$$(1) \quad 80 = -15t + \frac{1}{2} \times 9.8 \times t^2$$

$$(1) \quad 4.9t^2 - 15t - 80 = 0$$

$$(1) \quad t = 5.85 \text{ s}$$

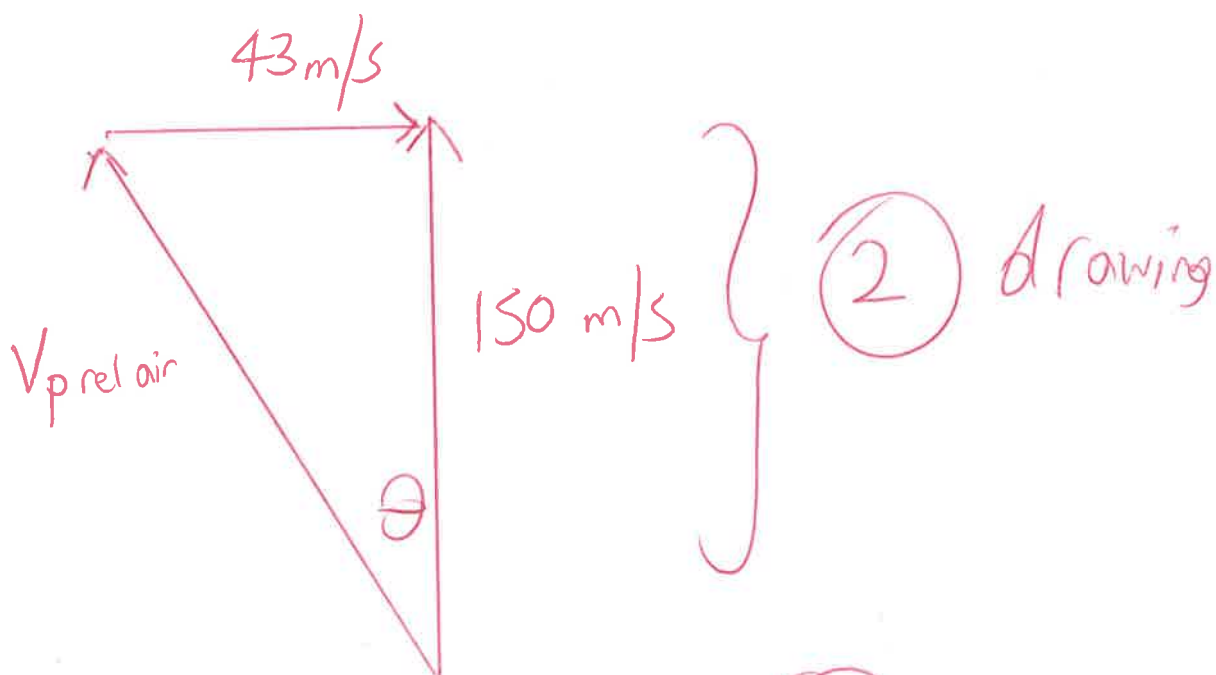
3

OR (1) time to top 1.53 s
 (1) time from $91.48 \text{ m} \rightarrow \text{sea}$ 4.32 s
 (1) Addition = 5.85 s

Question 17 (5 marks)**Marks**

A plane at Airport A needs to fly to Airport B, which is 270 km North, in 30 minutes. At the time of the flight, there is a constant wind blowing at 43 ms^{-1} East.

By drawing a vector diagram, determine the direction of flight of the plane and the speed of the plane through the air.



$$\tan \theta = \frac{43}{150} \quad \theta = 16^\circ \quad (1)$$

5

$$V = 156 \text{ m/s} = 561.7 \text{ km/h}$$

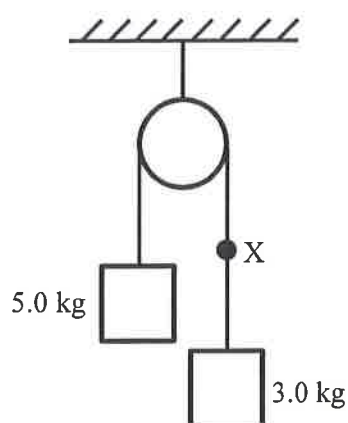
(1) (1)

Class

Name

Question 18 (4 marks)**Marks**

A simple vertical Atwood machine consists of a 3.0 kg mass and a 5.0 kg mass supported from a frictionless pulley by a light inextensible string as shown in the diagram below.



- (a) Calculate the magnitude of the acceleration of the two masses.

$$a = \frac{2g}{8} = \frac{9}{4}$$

$$2.45 \text{ m/s}^2$$

2

- (b) Calculate the magnitude of the tension force in the string at X.

$$T - 3g = 3a$$

$$T = 3(a + g)$$

$$= 36.75 \text{ N}$$

2

$$T = 3(a + g)$$

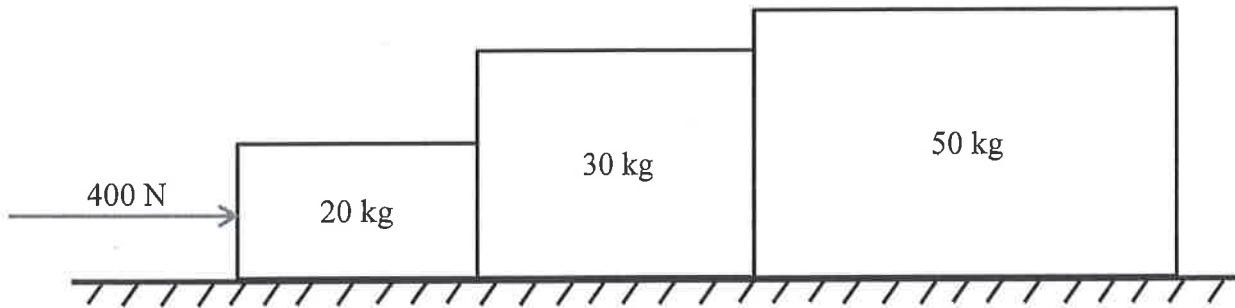
4

Class

Name

Question 19 (4 marks)**Marks**

Three blocks are pushed to the right on a smooth (ie frictionless) horizontal surface as shown below.



- (a) Calculate the magnitude of the acceleration of the masses.

$$a = 4 \text{ m/s}^2$$

1

- (b) Determine the magnitude of the net force on the 30 kg mass.

$$ma = 30 \times 4 = 120 \text{ N}$$

1

- (c) Calculate the magnitude of the force of the 30 kg mass on the 20 kg mass.

$$400 - F = 20a \quad (1)$$

$$F = 400 - 20a$$

$$= 320 \text{ N}$$

(1)

2

4

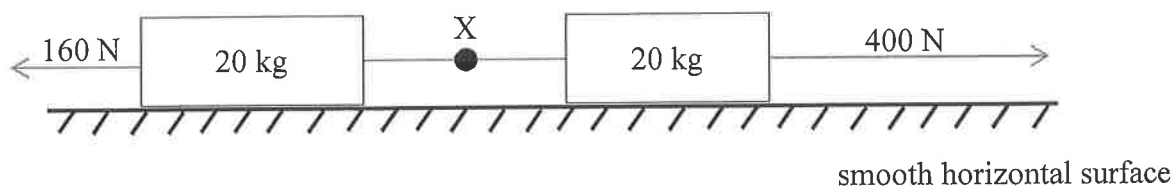
Class

 Name

Marks

Question 20 (4 marks)

Two masses are pulled along a smooth, horizontal surface by two horizontal forces as shown below.



- (a) Determine the magnitude of the acceleration of the masses.

$$a = \frac{240}{40} = 6 \text{ m/s}^2$$

1

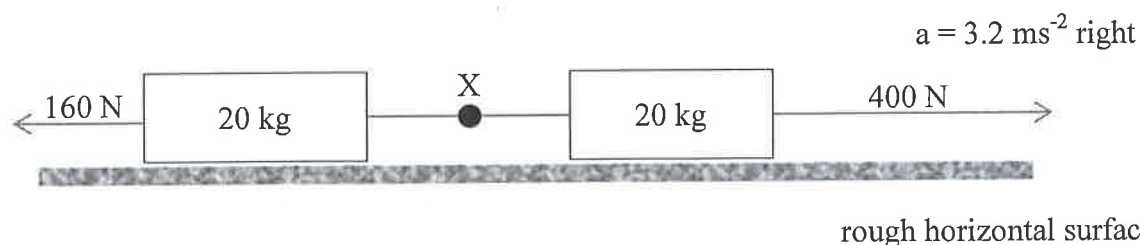
- (b) Determine the magnitude of the tension force in the rope at X.

$$T - 160 = 20a$$

$$T = 280 \text{ N}$$

1

- (c) If the same arrangement of masses is now placed on a rough surface, it is found that the acceleration is 3.2 ms^{-2} to the right as shown in the following diagram.



Calculate the magnitude of the force of friction acting on each mass (assuming that it is the same for each mass).

$$240 - 2f = 40 \times 3.2 \quad \textcircled{1}$$

$$f = 56 \text{ N} \quad \textcircled{1}$$

2

AAN

Class

CRIB

Name

Marks

Question 21 (2 marks)

A man of mass 80 kg is climbing up a rope as shown below.



If the acceleration of the man is 1.0 ms^{-2} upwards, determine the magnitude of the tension force in the rope.

$$T - Mg = ma$$

$$(\Sigma F = 80 \times 1 = 80 \text{ N})$$

(1)

2

$$\therefore T_R = 80 \times 10.8$$

$$= \underline{864 \text{ N}}$$

(2)

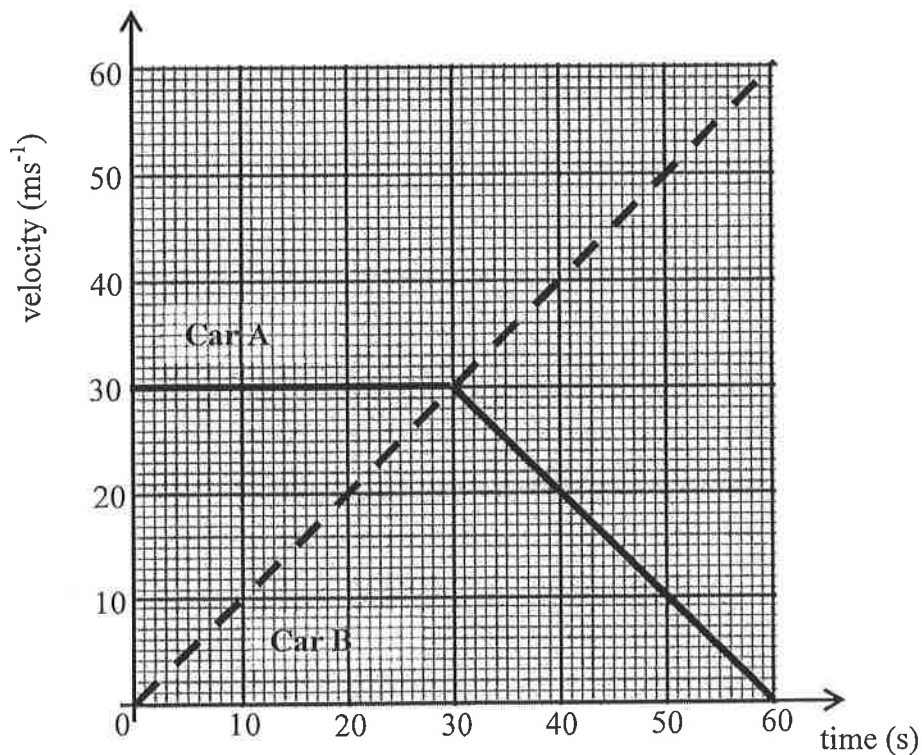
Class

 Name

Marks

Question 22 (5 marks)

The graph shown below represents the velocity of two cars racing along a straight road over a period of 60 s. The race begins at $t = 0$ s when the two cars are side by side.



- (a) Determine the speed of car A at 10 s.

30 m/s — ① 1

- (b) Calculate the magnitude of the acceleration of car B at 20 s.

= 60/60 = 1 m/s² — ① 1

Question 22 continued on next page.

Class

 Name

Question 22 continued.

Marks

- (c) Calculate the time when car B catches up with Car A again.

At $t = 30s$ $v_A = 30 \times 30 = 900m$ } ①
 $v_B = \frac{1}{2} \times 30 \times 30 = 450m$

3

From 30s:

$$v_B = 450 + 30t + \frac{t^2}{2} \quad v_A = 900 + 30t - \frac{t^2}{2}$$

→
Combine

$$450 + 30t + \frac{t^2}{2} = 900 + 30t - \frac{t^2}{2}$$

$$\therefore t^2 = 450$$

$$\therefore t = 21.2s$$

$$\therefore t_{\text{time}} = 21.2 + 30$$

$$= \underline{\underline{51.2s}} \quad \text{--- ①}$$

Class

 Name
Question 23 (7 marks)**Marks**

On a distant planet, an alien pupil performs an experiment to measure the acceleration due to gravity. He drops (i.e. from rest) a ball from various heights and measures the time the ball takes to hit the ground.

The equation relating the height dropped by the ball from rest and time (t) is given by:

$$\text{height} = \frac{1}{2}gt^2$$

where g = acceleration due to gravity.

He obtains the following data:

Height (m)	Time (s)	<i>Time² (s²)</i>
0.5	0.24	<i>0.058</i>
1.0	0.41	<i>0.168</i>
1.5	0.45	<i>0.203</i>
2.0	0.55	<i>0.303</i>
2.5	0.61	<i>0.372</i>
3.0	0.65	<i>0.423</i>

Question 23 continued on next page.

(NOT MARKED)

Conf.

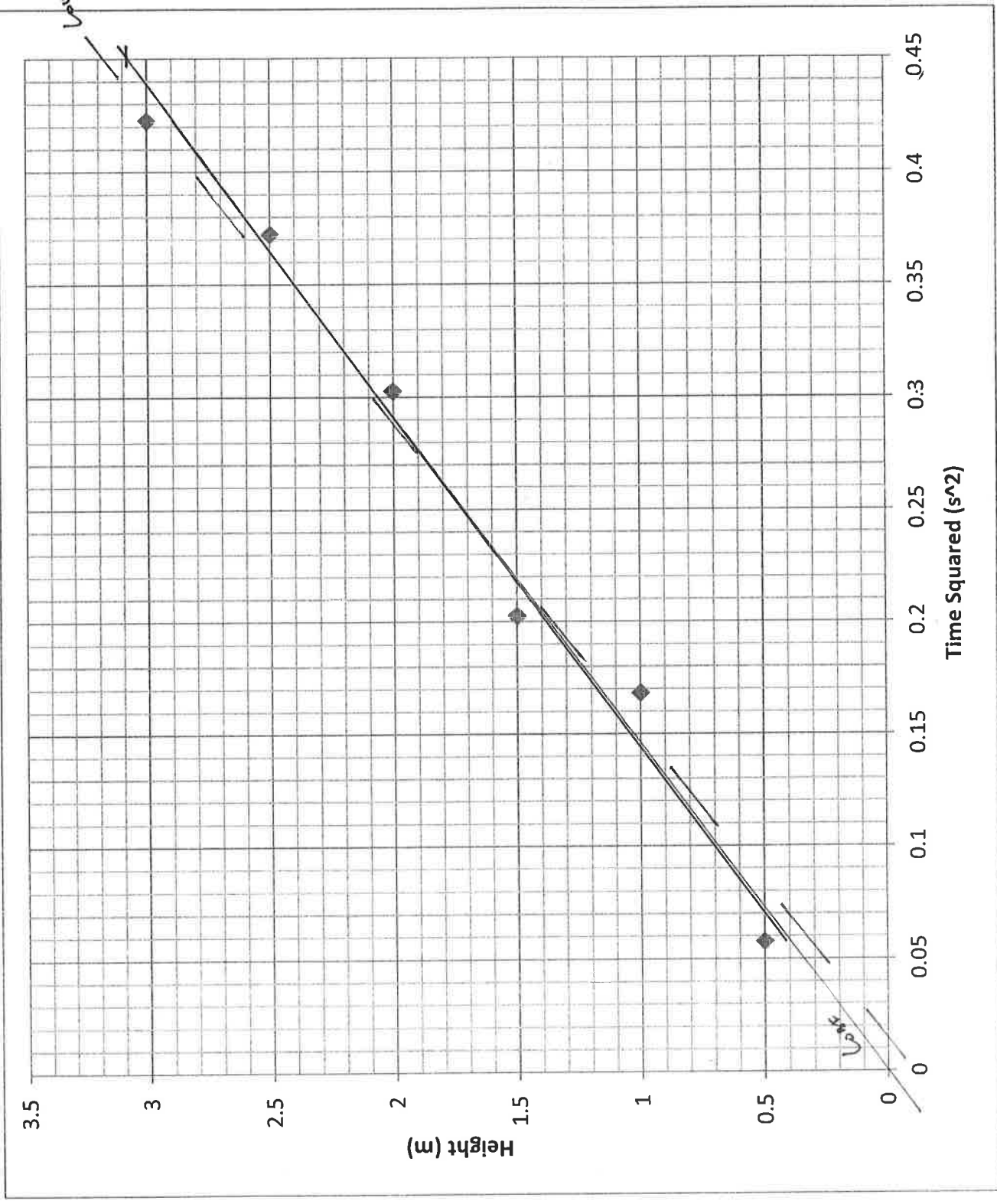
$$\text{Gradient} = \frac{1}{2}g$$

$$= \frac{3.08}{0.45} = 6.84$$

$$\therefore a = 13.7 \text{ m/s}^2$$

$$13.7 \pm 1.0$$

3m



Conf

$$\frac{7.18}{0.45} = 0.014$$

$$14.6$$

$$14.6$$

See next page for Marking Scheme.

Class

 Name

Question 23 continued.

Marks

- (b) Calculate the value for the acceleration due to gravity obtained by the alien pupil.

$$g = 2 \times \text{grad} \quad (\text{For Height v Time}^2 \text{ Graphs}) \quad 3$$

$$\text{grad} = \frac{\text{rise}}{\text{run}} = \frac{3.08}{0.45} = 6.84 \quad \text{--- (1)}$$

$$\therefore g = 2 \times 6.84 = \underline{\underline{13.7 \text{ m/s}^2}} \quad \text{--- (2)}$$

(Accepted $\pm 1.0 \text{ m/s}^2$)

↳ (2) if outside mis Range.

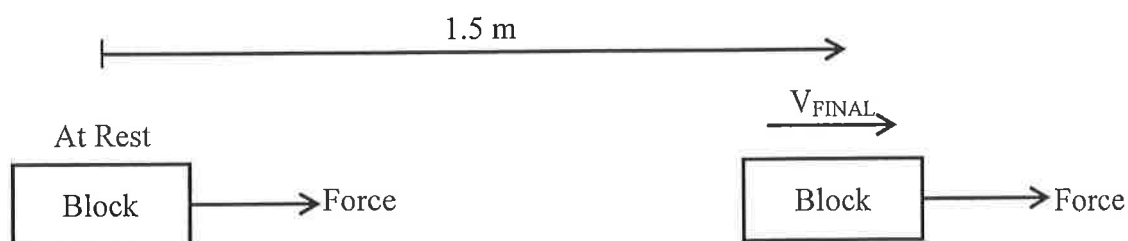
From Previous Page: For 4 marks ...

- STRAIGHT LINE GRAPH (ie $h \propto t^2$, $t^2 \propto h$ etc etc)
- Points Correct
- Axes LABELLED
- Axes with CORRECT UNITS
- SCALE
- CURVES
- LINE OF BEST FIT.

(-1) For Every Error.

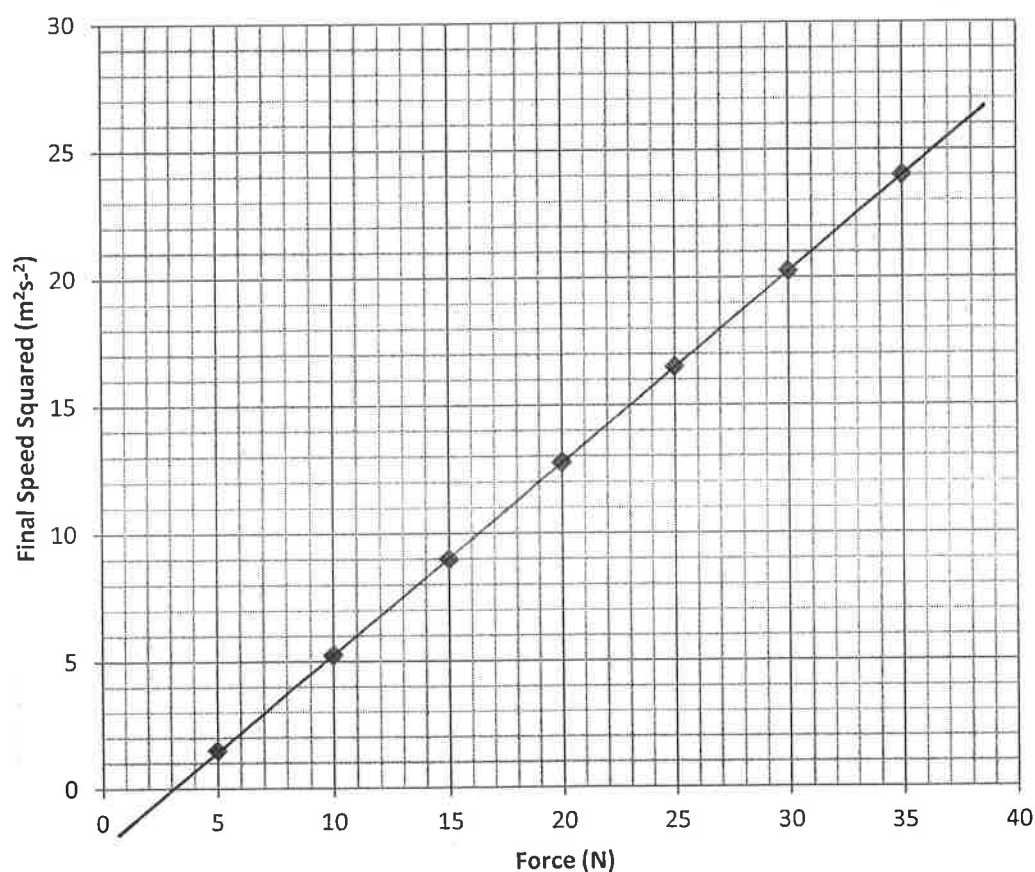
ClassName**Question 24** (3 marks)**Marks**

The diagram below shows an experiment performed by a boy in a Physics lesson.



The boy pulls a block, from rest, along the desk over a distance of 1.5 m with a constant force. A light gate measures the instantaneous speed of the block as it passes the 1.5 m mark.

The boy believes that a constant force of friction acts on the block during his experiments. He varies the force with which he pulls the block and obtains the following graph:



Question 24 continued on next page.

Class

Name

Question 24 continued.

Marks

Use the graph to determine:

- (a) the force of friction acting.

= INTERCEPT ON GRAPH

= 3

1

(ACCEPTED 2.5 - 3.5)

- (b) the mass of the block.

$$a = \frac{F - F_f}{M}$$

2

$$\therefore V^2 = u^2 + 2ar$$

$$V^2 = \frac{2(F - 3) \cdot 1.5}{M}$$

$$V^2 = \frac{3F}{M} - \frac{4.5}{M} \Rightarrow \text{Since}$$

$$\text{GRADIENT} = \frac{3}{M}$$

$$\text{GRADIENT} = \frac{\text{RISE}}{\text{RUN}} = 0.75 = \frac{3}{M}$$

$$M = 4 \text{ kg}$$

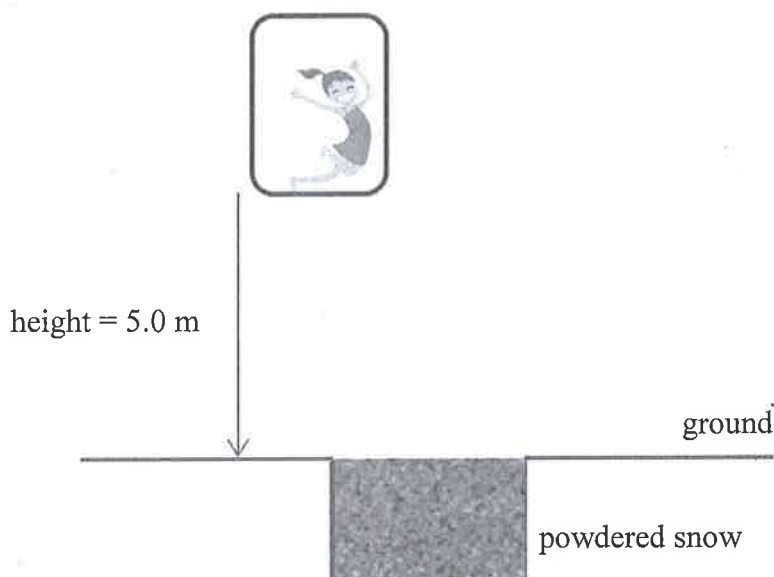
OR BY SUITABLE SUBSTITUTION.

Class

Name

Question 25 (6 marks)**Marks**

Natalia, of mass 50 kg, drops from a window 5.0 m above the ground as shown in the following diagram.



- (a) Determine Natalia's gravitational potential energy when she is 5.0 m above the ground.

$$GPE = mgh = 50 \times 9.8 \times 5 = 2450 \text{ J}$$

1

① correct answer

- (b) Determine the magnitude of her velocity when she first hits the ground.

$$KE = \frac{1}{2}mv^2 = 2450$$

Carry forward if required 2

$$v = \sqrt{\frac{2 \times 2450}{50}}$$

$$= 9.9 \text{ m/s.}$$

① Equate

Energy
or $v^2 = u^2 + 2as$

① correct answer

Question 25 continued on next page.

Class

Name

Question 25 continued.

Marks

- (c) The ground onto which Natalia falls is made of powdered snow. The snow applies a constant braking force upwards of 5.20×10^3 N on her.

Calculate the maximum distance Natalia will sink into the snow before stopping.

Energy

Simple $2450 = F + r$ ①

$$r = \frac{2450}{5200} = 0.471 \text{ m} = 47 \text{ cm} \cdot ①$$

Does not take into account additional energy if no velocity

Extended $2450 + mgr = Fr$ (will still sink if $v=0$)

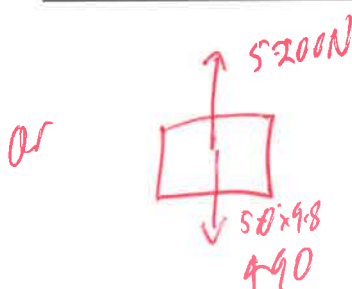
$$r(F - mg) = 2450$$

$$r = \frac{2450}{5200 + 50 \times 9.8}$$

$$= 0.52 \text{ m} = 52 \text{ cm} \cdot ①$$

3

Forces



Balance ①

$$\Sigma F = 5200 - 490 = 4710 = ma = 50a$$

$$a = -94.2 \text{ m/s}^2$$

①

$$v = 0, a = -9.9$$

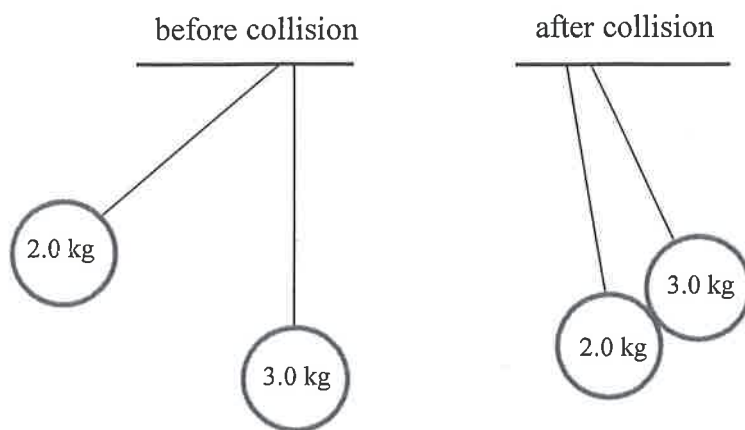
$$r = \frac{v^2 - u^2}{2a} = \frac{-9.9^2}{2(-94.2)} = 0.52 \text{ m} = 52 \text{ cm} \cdot ①$$

Class

Name

Question 26 (5 marks)**Marks**

A 2.0 kg pendulum moving at 4.0 ms^{-1} collides with a stationary 3.0 kg pendulum and then they stick together.



- (a) Determine the magnitude of the momentum of the 2.0 kg pendulum immediately before the collision.

$$p = mv = 2 \times 4 = 8 \text{ kg m/s}$$

$$8 \text{ N s}$$

1

magnitude only

- (b) Determine the velocity of the combined pendulums immediately after the collision.

$$\sum p_i = \sum p_f$$

$$(2 + 3)v = 2 \times 4 = 8$$

$$v = \frac{8}{5} = 1.6 \text{ m/s}$$

2

Question 26 continued on next page.

Class

Name

Question 26 continued.

Marks

- (c) Ignoring the diameters of the pendulums, determine the vertical height the combined pendulums reach after the collision.

$$KE \text{ at bottom} = GPE \text{ at maximum height}$$

$$\frac{1}{2}(2+3)(1.6)^2 = (2+3)gh$$

$$h = \frac{1.6^2}{2 \times 9.8} = 0.13 \text{ m (or 13 cm)}$$

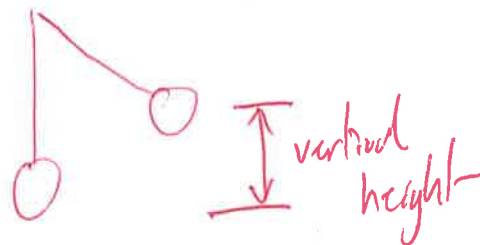
2

The KE before cannot be used because the collision is not elastic. Only

① $KE = GPE$

or

① Calculate $KE = \frac{1}{2}5(1.6)^2 = 6.4 \text{ J}$



$$\text{Work} = \Delta KE = mgh = \frac{1}{2}mv^2 - \frac{1}{2}mu^2 = mgh$$

$$v^2 - u^2 = 2gh$$

So $v^2 = u^2 + 2gh$
works in this specific example only - bad physics

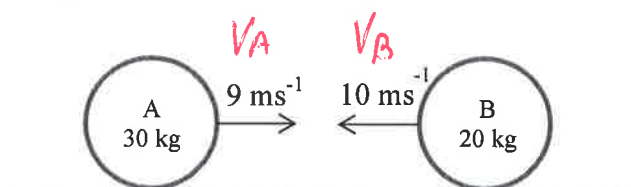
Class

Name

Marks

Question 27 (8 marks)

Two objects, A and B, approach each other and then collide. After the collision object A is travelling at half the **speed** of object B.



- (a) Calculate the total momentum of this system before the collision.

$$\Sigma p_i = 30 \times 9 + 20 \times (-10)$$

$$= 70 \text{ kg m/s} \text{ right or positive number}$$

① - Summing momenta ① magnitude and direction

- (b) (i) Find all possible solutions for the final velocities of object A and object B.

Conservation of Momentum. $\Sigma p_{\text{before}} = \Sigma p_{\text{after}}$
 $70 = 30V_A + 20V_B$ (or $7 = 3V_A + 2V_B$)

V_A is half the speed of V_B , $V_B = \pm 2V_A$ or $V_B = \pm \frac{V_A}{2}$

Case 1 $V_B = 2V_A$ same direction Case 2 $V_B = -2V_A$

$$7 = 3V_A + 2(2V_A) = 7V_A$$

$$V_A = 1 \text{ m/s}, V_B = 2V_A = 2 \text{ m/s}$$

(A) $\xrightarrow{1 \text{ m/s}}$ (B) $\xrightarrow{2 \text{ m/s}}$

$$V_A = 1 \text{ m/s}, V_B = 2 \text{ m/s}$$

$$7 = 3V_A + 2(-2V_A)$$

$$7 = 3V_A - 4V_A, V_A = -7 \text{ m/s}, V_B = -2(-7) = 14 \text{ m/s}$$

(A) $\xleftarrow{7 \text{ m/s}}$ (B) $\xrightarrow{14 \text{ m/s}}$

$$V_A = -7 \text{ m/s}, V_B = 14 \text{ m/s}$$

• Collision is not elastic

• These are the only 2 valid solutions.

• no elastic solutions exist for $V_B = \pm 2V_A$ where speed of A is half speed of B.

- (ii) Identify which solution you think is most likely. Justify your answer.

$$V_A = 1 \text{ m/s} \text{ \& } V_B = 2 \text{ m/s} \text{ is the actual solution.}$$

Total KE is lost ($2215 \text{ J} \rightarrow 55 \text{ J}$), other solution

$V_A = -7 \text{ m/s}, V_B = 14 \text{ m/s}$ KE is gained which violates conservation of energy
 $(2215 \rightarrow 2695 \text{ J})$

ADDITIONAL CRIB NOTES FOR Q27

Q27bi)

Successfully forming an equation of conservation of momentum (with substitution) ie $70 = 30 v_A + 20 v_B$	1
Substitution of one or both of $v_B = \pm 2v_A$	1
Correctly calculates one solution set of v_A & v_B	1
Both correct valid solutions	1

Solution 1

$$v_A = 1, v_B = 2$$

Solution 2

$$v_A = -7, v_B = 14$$

Comments

- There is no indication in the question that the collision is elastic. So conservation of kinetic energy cannot be used
- Solutions occur in pairs of velocities for objects A & B. Answers such as $v_A = \pm 1$, $v_B = \pm 2$ do not indicate paired solutions
- Many boys completely ignored that one object is moving twice the speed of the other. This made identifying wrong solutions easy.
- Conservation of momentum is true for all collisions. Also total momentum = +70 so answers that assume $v_A = -1$, $v_B = -2$ is valid are simply wrong.
- Many boys are using 1 and 2 as subscripts and are not properly distinguishing which objects A or B velocity they are talking about
- $U_A - U_B = V_A - V_B$ is only true for elastic collisions and not valid

Q27bii)

Despite appearances the question is not really asking your personal opinion. You have to justify your answer using some form of Physics.

There is no mark for identifying a correct answer, all marks were for justifying your answer.

a valid explanation that makes a clear distinction between 2 valid answers. (sorry, having only one answer means no marks – the question tells you there is more than one answer. Unfortunately making comparisons when you have invalid answers leads to nonsensical justifications that cannot be accepted. Therefore carried forward errors could not be accepted and some papers were remarked accordingly. Neither solution is elastic, and therefore an assumption that it is “not the elastic one” is not valid, however comparisons with total energy loss or change in KE or are more valid.	1
Identifies (mathematically) that the kinetic energy (or combined velocities) is greater for one solution and without any addition of energy these solutions are invalid.	1

$$\text{Total KE (Before)} = 0.5 \cdot 30 \cdot 9^2 + 0.5 \cdot 20 \cdot (-10)^2 = 2215\text{J}$$

Solution 1

$$v_A = 1, v_B = 2$$

$$\text{Total KE (after)} = 0.5 \cdot 30 \cdot 1^2 + 0.5 \cdot 20 \cdot (2)^2 = 55\text{J}$$

This is an inelastic collision

Solution 2

$$v_A = -7, v_B = 14$$

$$\text{Total KE (after)} = 0.5 \cdot 30 \cdot (-7)^2 + 0.5 \cdot 20 \cdot (14)^2 = 2695\text{J}$$

Elastic collision means KE is conserved.

In this case there is more KE after than before.

It is not a valid collision as energy is not conserved (in this closed system).